RAJALAKSHMI ENGINEERING COLLEGE

THANDALAM, CHENNAI-602105

CR23532 – CYBER AND DIGITAL FORENSICS LABORATORY RECORD

Submitted by

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2116241901109

in partial fulfillment of the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

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NAME: SIVARANJANII. M

DATE: 28.7.26 ROLL No: 241901109

EXPERIMENT 1

STUDY OF COMPUTER FORENSICS AND DIFFERENT TOOLS USED FOR FORENSIC INVESTIGATION

AIM:

To study the concept of computer forensics and understand the different tools used for forensic investigation.

INTRODUCTION

Computer forensics is a branch of digital forensics that deals with identifying, collecting, preserving, and analyzing digital evidence from computers and other digital devices.

It is used to investigate cybercrimes, unauthorized access, data theft, and other security incidents. Digital evidence may include log files, deleted files, browser history, system activity, and other information stored on a computer.

THEORY

Digital forensic investigators examine computers to find evidence that helps them understand what happened during a security incident. Even when files are deleted, some information may remain and can sometimes be recovered.

The main steps in a digital forensic investigation are:

1. Identification: Identifying the devices and possible sources of digital evidence.
2. Collection: Collecting the relevant evidence carefully.
3. Preservation: Protecting the evidence from alteration or damage.
4. Examination: Examining the collected data using forensic tools.
5. Analysis: Interpreting the evidence to understand the incident.
6. Presentation: Preparing a report of the findings.

DIFFERENT TOOLS USED IN COMPUTER FORENSICS

1. EnCase Forensic

EnCase is a commercial forensic tool used to examine digital evidence, investigate computer incidents, and prepare forensic reports.

2. FTK (Forensic Toolkit)

FTK is a digital forensic tool used to analyze digital evidence, search files, and investigate information stored on computers.

3. ProDiscover

ProDiscover is a forensic investigation tool used to examine computer storage devices and analyze digital evidence.

4. Autopsy

Autopsy is a digital forensic platform that provides a graphical interface for examining digital evidence. It can help investigators analyze files, recover certain deleted files, and examine user activity.

5. The Sleuth Kit (TSK)

The Sleuth Kit is a collection of command-line tools used to examine file systems and analyze digital evidence.

6. FTK Imager

FTK Imager is used to create forensic images of storage devices and examine their contents. A forensic image is a copy of digital data created for investigation.

7. USBDeview

USBDeview is a utility that displays information about USB devices connected to a Windows computer, including previously connected devices.

PROCEDURE:

Studied the basic concepts and importance of computer forensics.

1. Identified the major stages of a digital forensic investigation.
2. Studied the functions of different forensic tools, including EnCase, FTK, ProDiscover, Autopsy, The Sleuth Kit, FTK Imager, and USBDeview.
3. Understood how these tools help investigators examine digital evidence and investigate security incidents.

RESULT

Thus, the concept of computer forensics, the stages of digital forensic investigation, and the functions of different forensic tools were studied successfully.

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EXPERIMENT 2

LIVE FORENSICS CASE INVESTIGATION USING AUTOPSY

AIM

To investigate digital evidence using Autopsy and examine files, metadata, and deleted-file information.

PROCEDURE

1. Open Autopsy and select Create New Case.

2. Enter the case details:

- Case Name: Live_Forensics
- Case Number: LF001
- Examiner Name: Your Name

3. Click Finish to create the case.

4. Select Add Data Source.

5. Choose Local Disk and select the authorized test computer's C: drive or the available forensic image.

6. Select the required ingest modules, such as Recent Activity, File Type Identification, Keyword Search and Hash Lookup.

7. Click Next or Finish to start the ingest process and wait for the analysis to complete.

8. Open Data Sources and examine the available user folders, such as Desktop, Documents, Downloads and Pictures.

9. Navigate to Results → Recent Activity to examine available browser activity and other recorded user activities.

10. Use Keyword Search to search for relevant terms, if the module is enabled and configured.

11. Open the Deleted Files section, if available, and examine the identified deleted files.

12. Select an important file and examine its details, such as file path, size, created time, modified time and accessed time, wherever available.

13. Record the relevant evidence and timestamps in the observation table.

14. Tag important evidence as an Interesting Item or Notable Item, depending on the available options.

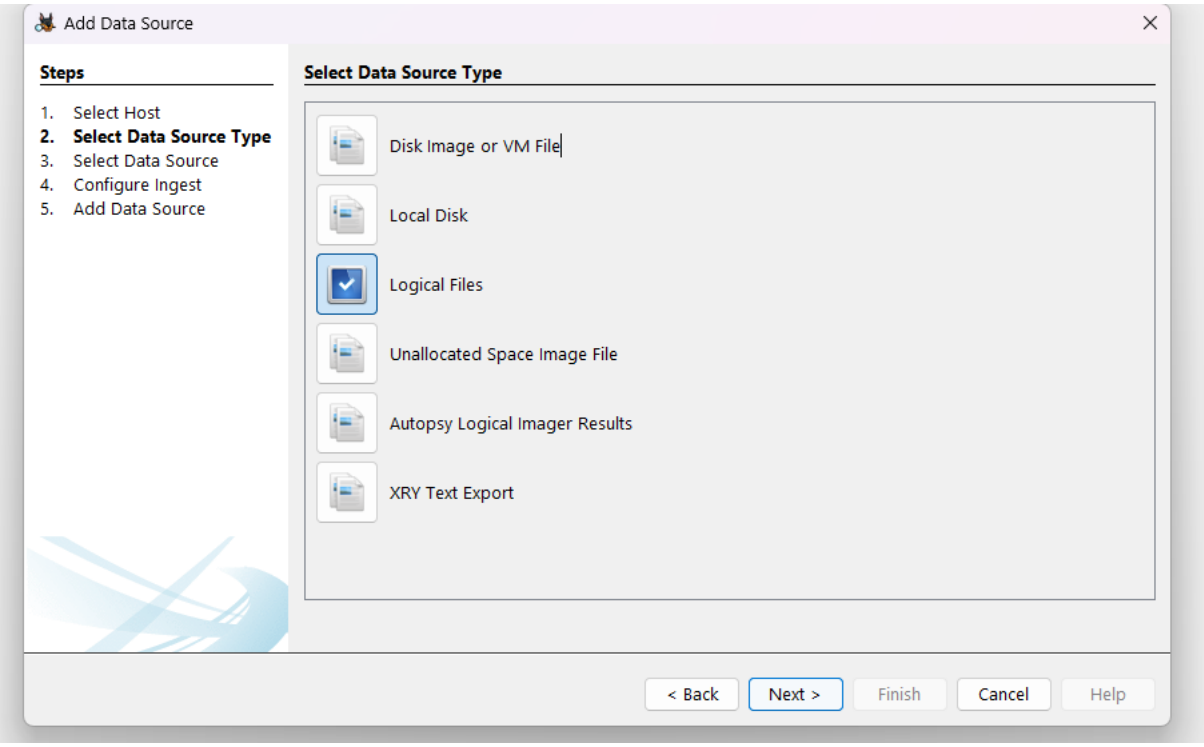
15. Generate an HTML report using Tools → Generate Report, if supported by the installed version.

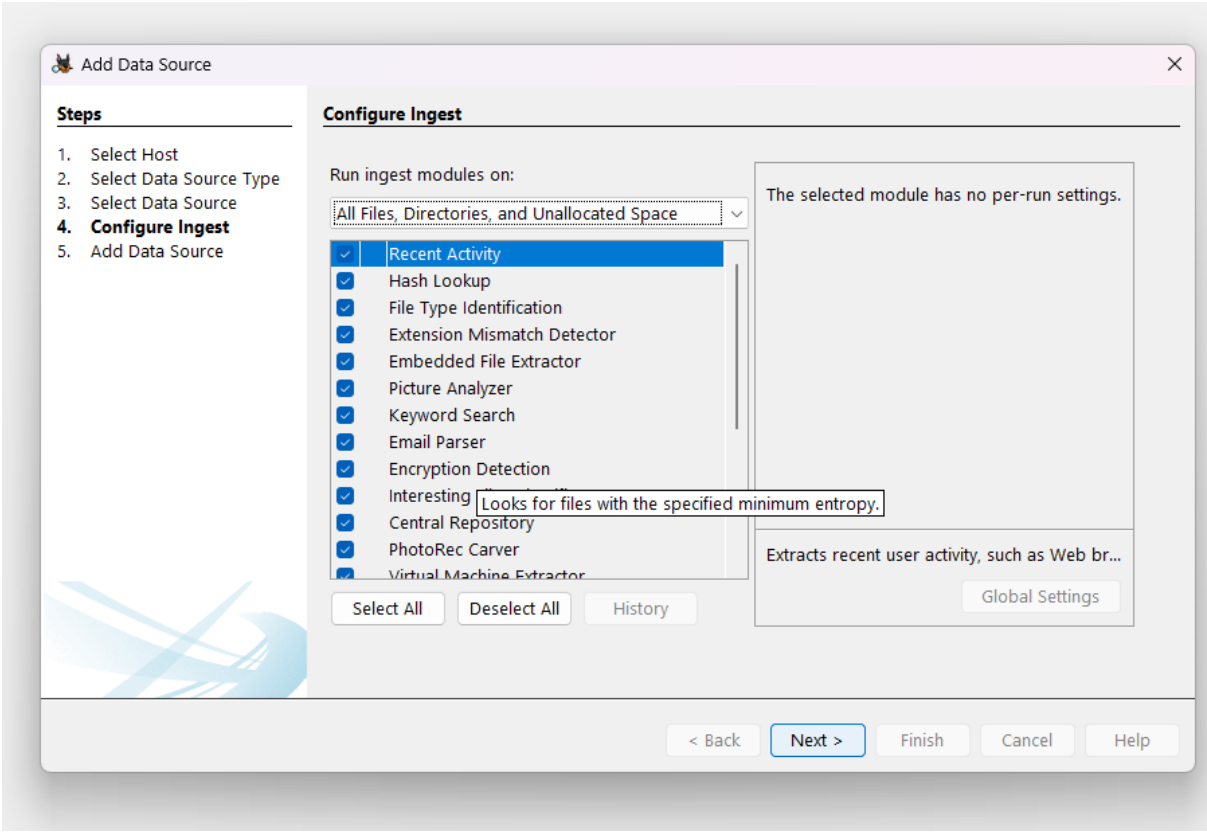
16. Open the generated report and verify the documented findings.

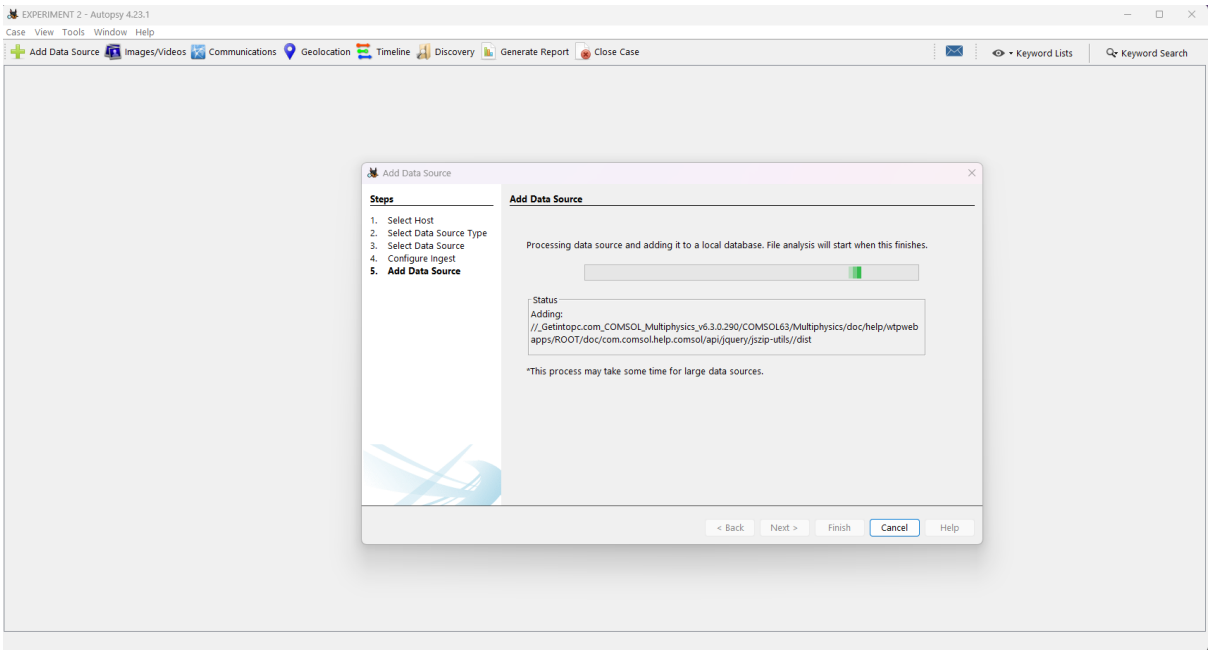
OBSERVATION

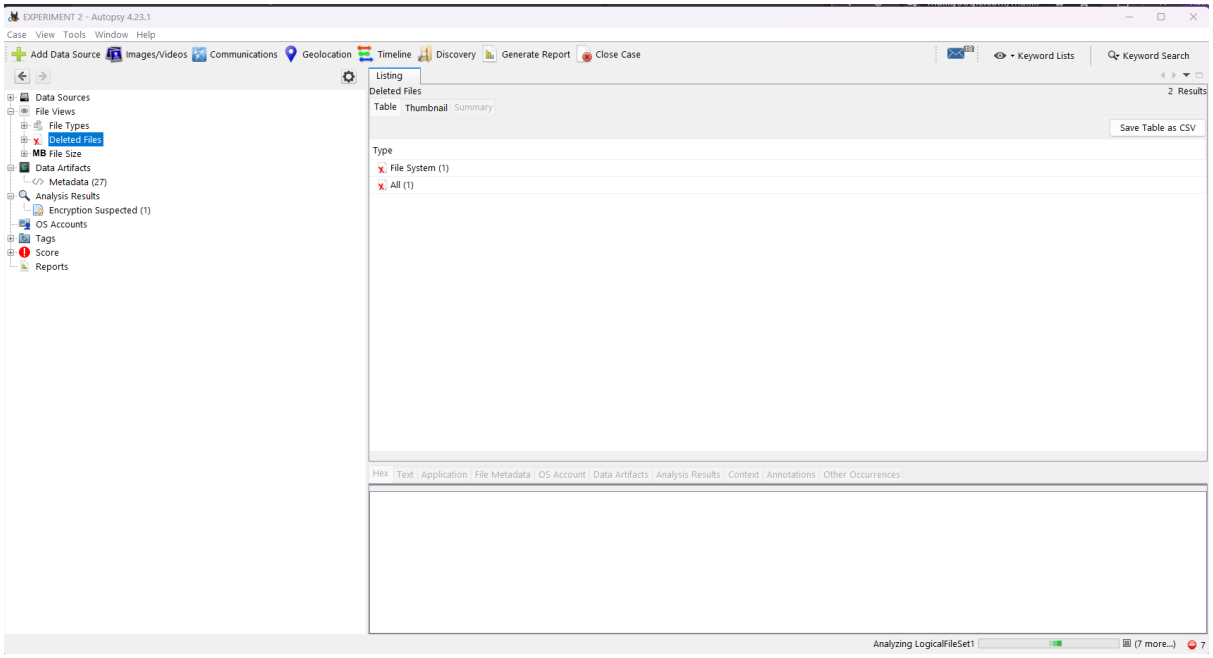
Autopsy processed the selected logical files and displayed forensic information such as file metadata, creation and modification dates, and deleted-file entries.

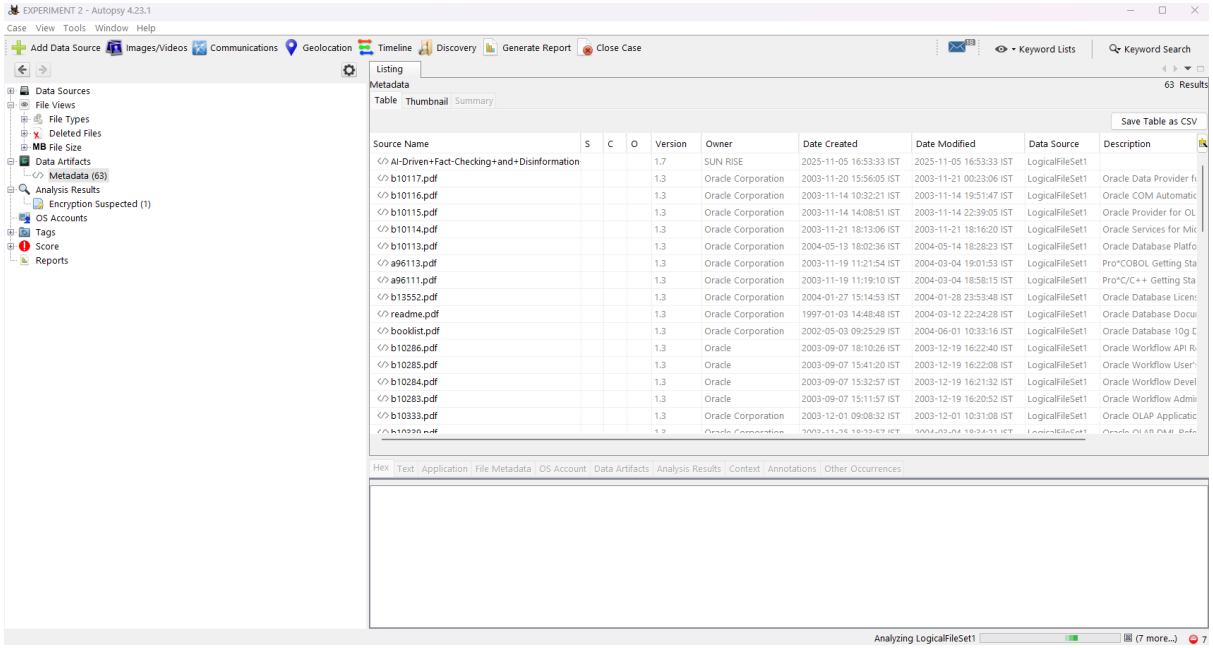
EVIDENCE EXAMINED	FINDING
User files	Files were identified in Desktop, Documents, Downloads and Pictures folders.
Browser history	Recent browsing activities were identified, where available.
Downloaded files	Downloaded files were identified and their locations were examined.
Keyword search	Keyword search results were examined for matching terms.
Deleted files	Deleted files were identified, if available.
File timestamps	File creation and modification timestamps were examined using Autopsy.
Tagged evidence	Important evidence items were tagged successfully.




Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Configure Ingest
Run ingest modules on:
All Files, Directories, and Unallocated Space
Recent Activity
Hash Lookup
File Type Identification
Extension Mismatch Detector
Embedded File Extractor
Picture Analyzer
Keyword Search
Email Parser
Encryption Detection
Interesting
Looks for files with the specified minimum entropy.
Central Repository
PhotoRec Carver
Select All
Deselect All
History
The selected module has no per-run settings.
Extracts recent user activity, such as Web br...
Global Settings
< Back
Next >
Finish
Cancel
Help


EXPERIMENT 2 - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Add Data Source
Processing data source and adding it to a local database. File analysis will start when this finishes.
Status
Adding:
//_GetintopC.com_COMSOL_Multiphysics_v6.3.0.290/COMSOL63/Multiphysics/doc/help/wtpwebapps/ROOT/doc/com.comsol.help.comsol/api/jquery/jszip-utils//dist
*This process may take some time for large data sources.
< Back
Next >
Finish
Cancel
Help





RESULT

Thus, digital evidence was examined using Autopsy, and the available file metadata and deleted-file information were observed.

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EXPERIMENT 3

RECOVER DELETED FILES USING FORENSICS TOOLS

AIM

To recover deleted files from a storage device using forensic tools and examine the recovered data.

PROCEDURE

1. Acquire Image of USB Drive
    - Create a forensic image of the USB drive using FTK Imager or the dd command.
    - Ensure write-blocking to maintain data integrity.
2. Load Evidence in Tool
    - Open Autopsy or FTK Imager.
    - Add the USB image file (E01, RAW or DD format) as evidence.
3. Scan for Deleted Files
    - Allow the tool to process the file system.
    - Check the Unallocated Space and Deleted Files sections.
4. Recover Files
    - Identify deleted files, such as .docx, .pdf and .jpg.
    - Export them to an evidence folder.
5. Verify Integrity
    - Generate MD5 or SHA-1 hash values for the recovered files.
    - Compare them with the original files, if available, to check integrity.
6. Document Findings
    - Prepare a forensic report containing:
        - File names and types

- File size and location
- Recovery status
- Hash values

STEP-BY-STEP PROCEDURE (Using Autopsy)

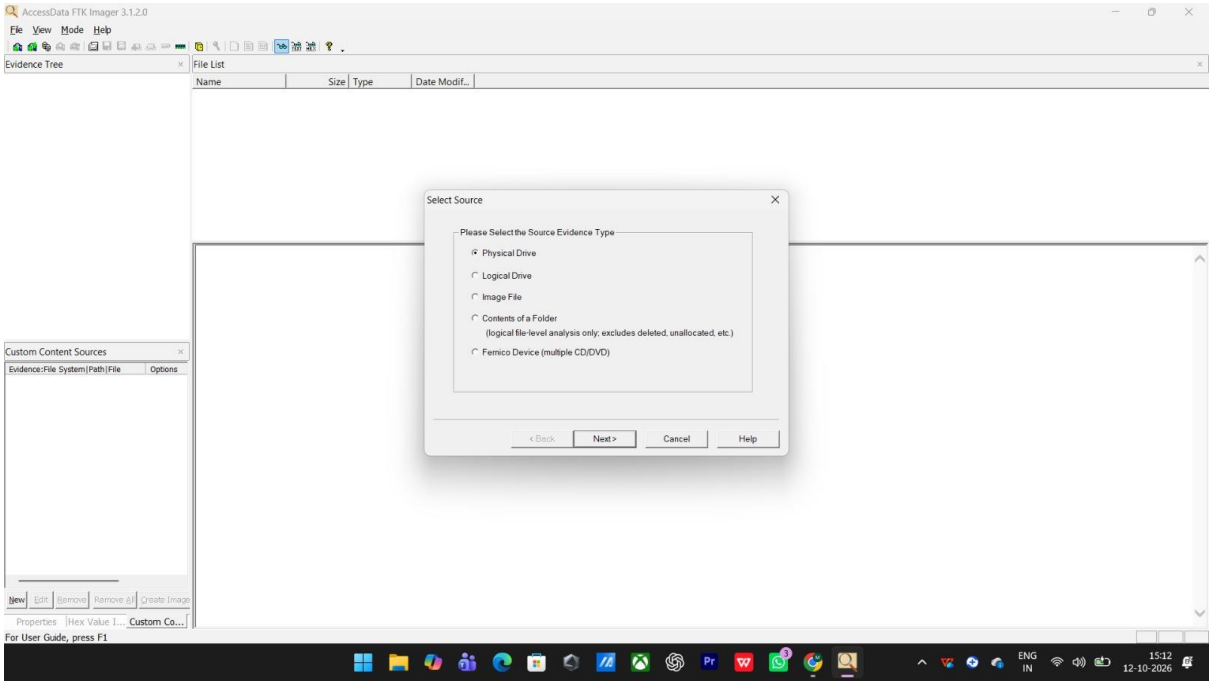
1. Launch Autopsy and create a new case.
2. Enter the case name as USB Deleted File Recovery.
3. Add the USB image file as the data source.
4. Configure ingest modules and enable File Type Identification and Deleted Files.
5. Start processing and wait for the analysis to complete.
6. Navigate to Views → Deleted Files.
7. Select the required files and use the available extraction or recovery option to save them in a secure evidence folder.
8. Use Autopsy's reporting feature to generate an HTML or PDF report.

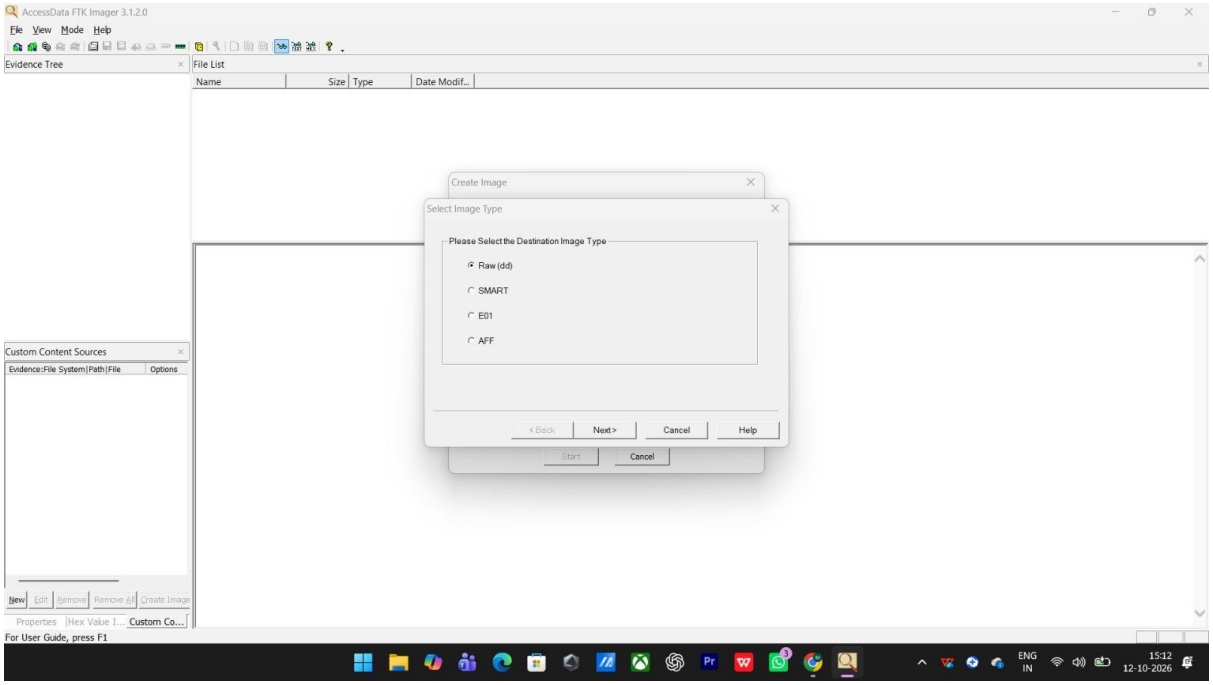
OBSERVATION

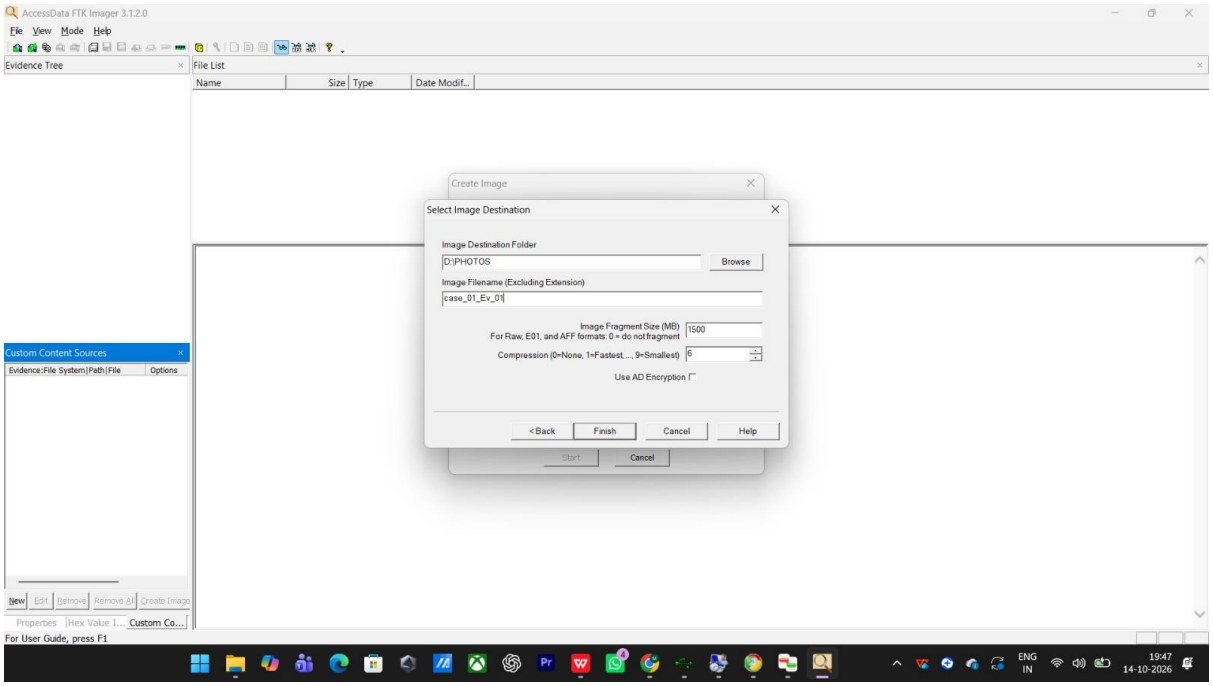
The USB image was loaded into the forensic tool and scanned for deleted files. Deleted files, if available, were identified and examined. The recoverable files were exported to an evidence folder, and their details were recorded for forensic analysis.

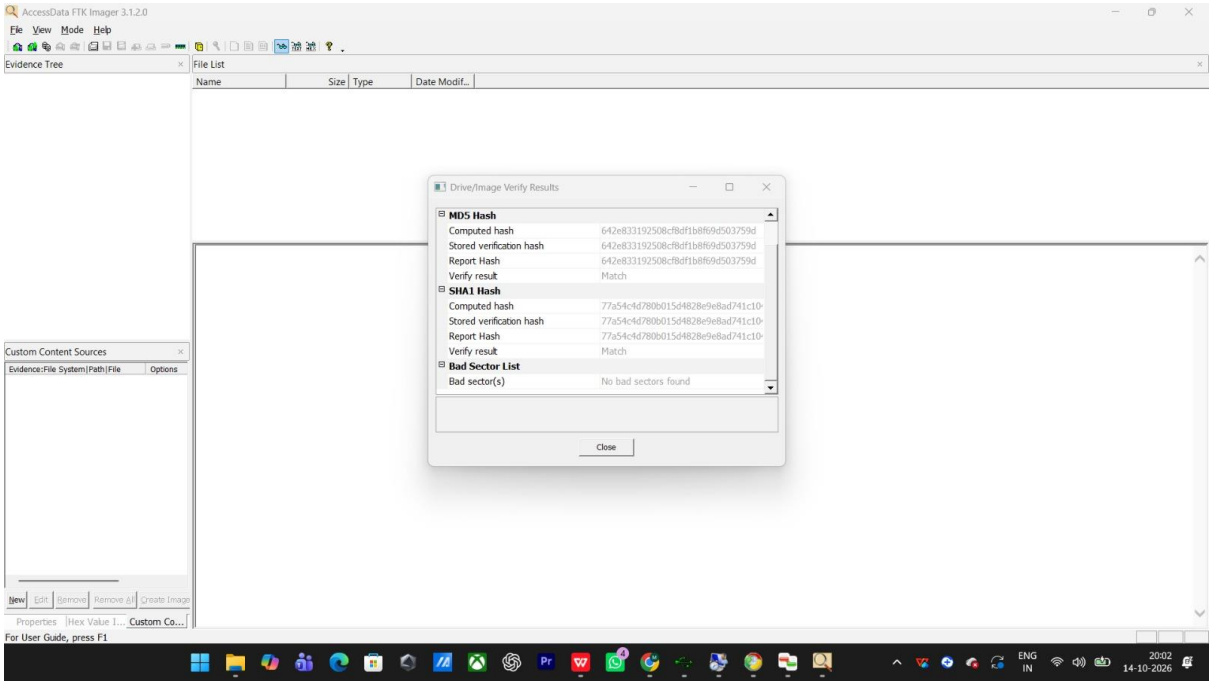
S.No.	Evidence Examined	Observation
1	USB image	USB forensic image was loaded into Autopsy.
2	Deleted files	Deleted files were examined in the Deleted Files section.
3	File types	File types such as DOCX, PDF and JPG were checked, where available.
4	Recovered files	Identified recoverable files were exported to an evidence folder, if available.
5	Hash values	MD5 or SHA-1 hash values were generated for integrity verification, if performed.
6	Forensic report	Findings were documented in a forensic report.

OBSERVATION









RESULT

Deleted files were searched for and recovered using forensic recovery tools, and the recovered data was examined successfully.

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DATE:18.8.26 ROLL No.:241901109

EXPERIMENT 4

FIND LAST CONNECTED USB ON YOUR SYSTEM (USB FORENSICS)

AIM:

To identify the last connected USB device on the system using USBDeview and record its details for forensic investigation.

PROCEDURE:

STEP 1: DOWNLOAD USBDeview

1. Open a web browser.
2. Visit the NirSoft website.
3. Download the USBDeview ZIP file.

STEP 2: EXTRACT THE FILE

1. Locate the downloaded ZIP file.
2. Right-click and select Extract All
3. Open the extracted folder.
4. Find USBDeview.exe

STEP 3: Run USBDeview

1. Right-click on USBDeview.exe
2. Select Run as administrator
3. The USBDeview main window will open.

STEP 4: View USB Devices

1. Open USBDeview to view connected and previously used USB devices.
2. Check the device name, description, serial number, Vendor ID, Product ID, and last plug/unplug date.

STEP 5: Identify Last Connected USB

1. Check the Last Plug/Unplug Date column.
2. Compare the dates and identify the latest connection.
3. Record the device details in the observation table.

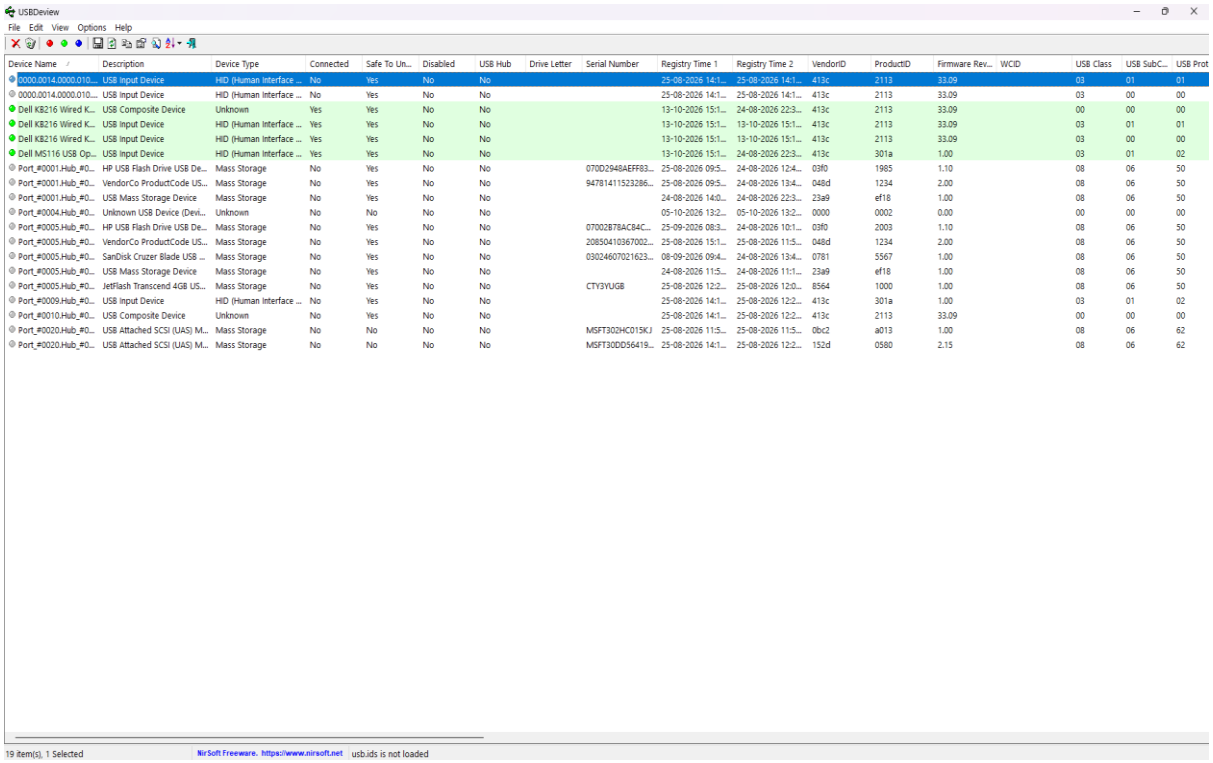
STEP 6: Record the Observation

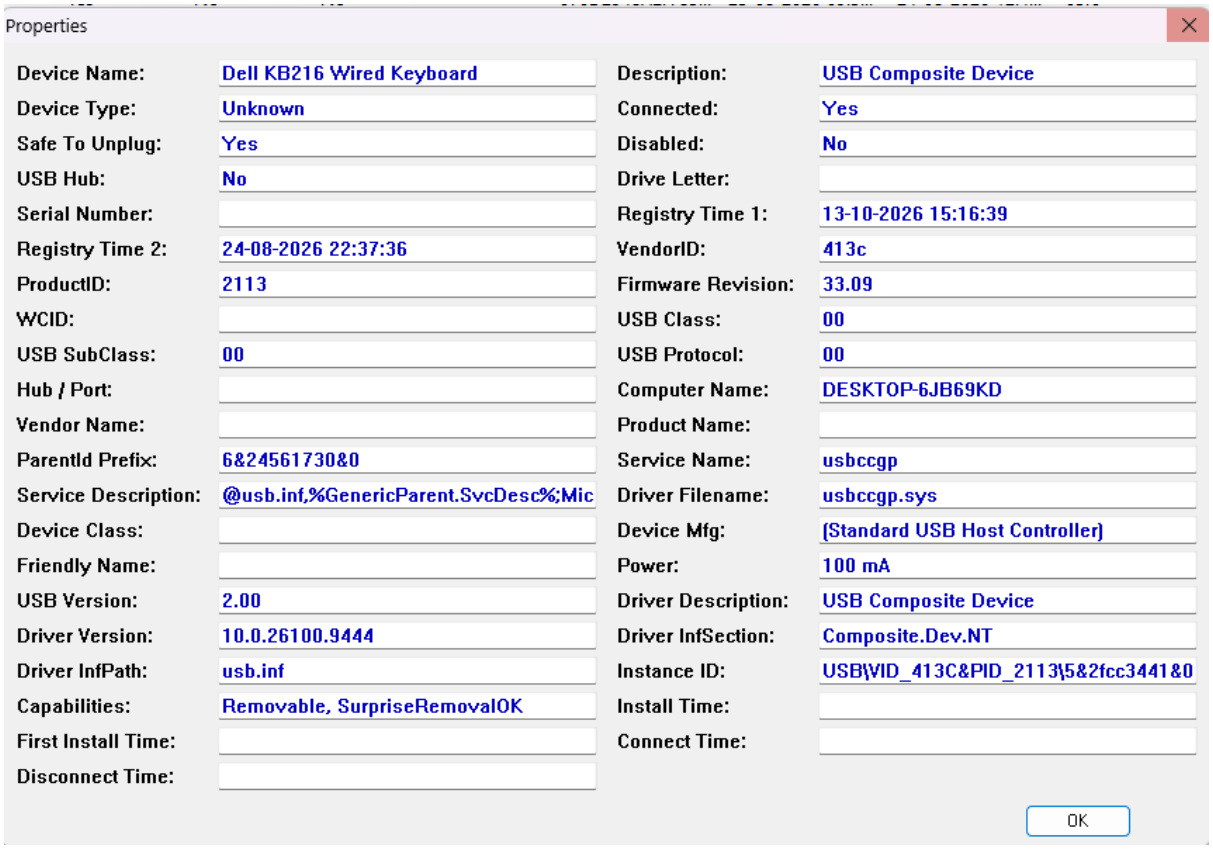
Enter the information displayed by USBDeview into the observation table in your record.

OBSERVATION:

The USBDeview tool was opened, and the USB devices previously connected to the system were displayed. The device details were examined, and the last connection date and time were recorded.

S.No.	DETAILS	OBSERVATION
1.	Device Name	SanDisk Cruzer Blade
2.	Description	USB Mass Storage Device
3.	Device Type	Mass Storage
4.	Serial Number	C4S3000106912108117
5.	Last Plug/Unplug Date	16-08-2026 04:32:10
6.	Vendor ID	0781
7.	Product ID	5567





RESULT:

Thus, the USBDeview tool was used to identify USB devices connected to the system, and the details of the last connected USB device were recorded successfully.

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DATE: 25.8.26 ROLL No.:241901109

EXPERIMENT 5

VIEW LAST ACTIVITY OF YOUR PC

AIM:

To identify and analyze the last activities performed on a computer using LastActivityView.

PROCEDURE:

STEP 1: Download LastActivityView

1. Open a web browser. Visit NirSoft website.
2. Download the LastActivityView ZIP file.

STEP 2: Extract the file

STEP 3: Run LastActivityView

1. Open LastActivityView.exe
2. Wait for the tool to collect available activity information.
3. The main window displays a list of recorded system and user activities.

STEP 4: View Recent Activities

1. Observe the list of activities displayed by the tool.
2. Check the activity name, date, time, and other available details.
3. Examine the events to understand the recent activity history of the computer.

STEP 5: Analyze Activity Details

1. Select an activity from the list. Check its record.
2. Compare the timestamps of different activities.
3. Identify relevant recent system events and user activities.

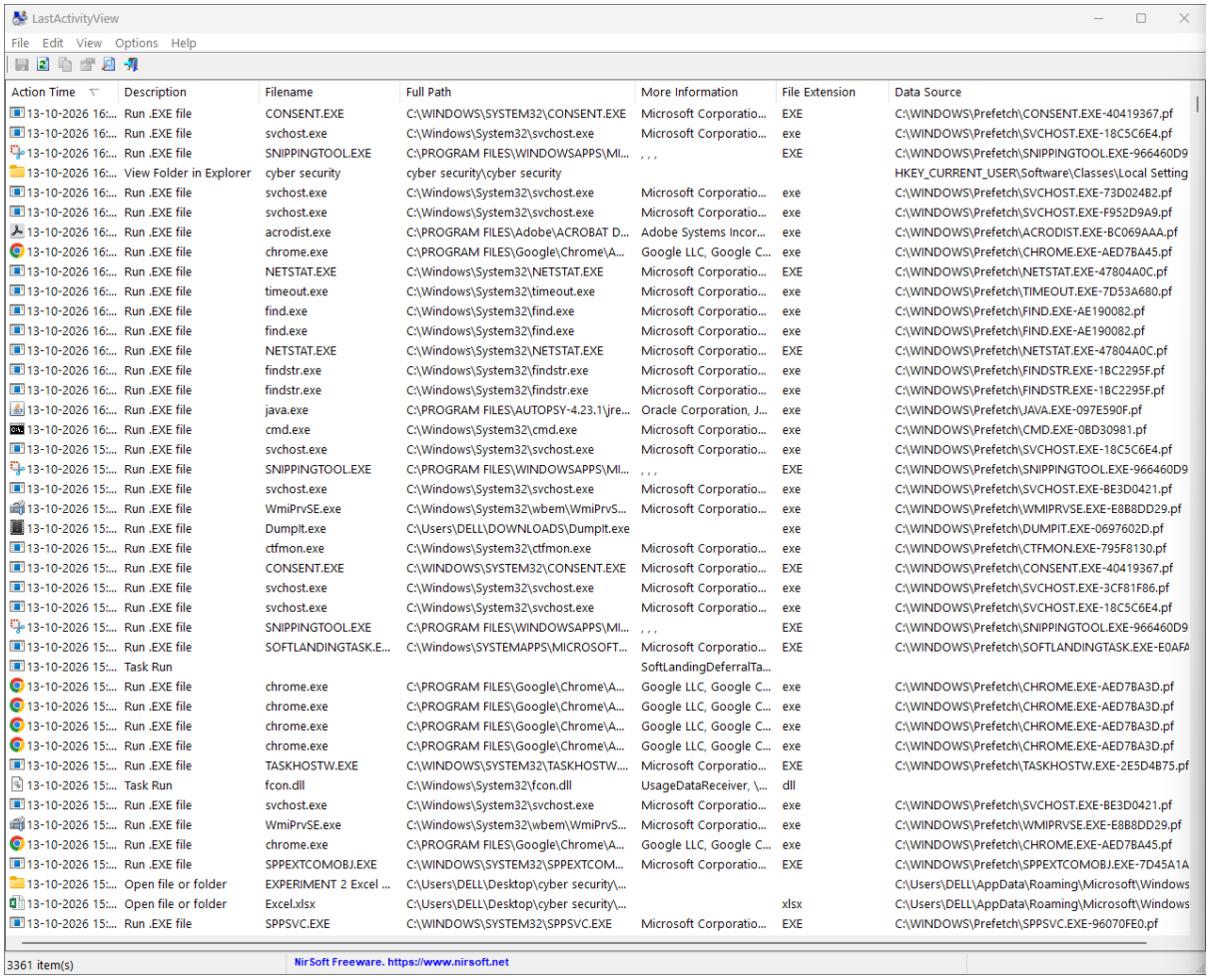
STEP 6: Record the Observation

OBSERVATION:

LastActivityView was used to display recent user activities and system events. The activity names and timestamps were examined and recorded for analysis.

S.No.	ACTIVITY DETAILS	OBSERVATION
1.	Activity Name	System Started
2.	Activity Time	20-09-2026, 9:00 AM
3.	Activity Name	File Accessed
4.	Activity Time	20-09-2026, 9:30 AM

LastActivityView was used to display recent user activities and system events. The activity names and timestamps were examined and recorded for analysis.



RESULT:

Thus, the recent activities of the PC were viewed and analyzed successfully using LastActivityView.

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DATE: 8.9.26 ROLL No.:241901109

EXPERIMENT 6

EXTRACTION OF BROWSER ARTIFACTS

AIM:

To extract and analyze browser artifacts such as history, cache, and cookies using forensic tools.

PROCEDURE:

STEP 1: Download Browser Forensic Tools

1. Download Chrome HistoryView and Chrome Cache View from the NirSoft website.

STEP 2: Run Chrome HistoryView

1. The tool reads the Google Chrome history database.
2. Observe the list of visited web pages.
3. Check the URL, page title, visit date/time, and visit count.

STEP 3: View Browser History

1. Examine the displayed website history.
2. Identify the recently visited websites.
3. Note the URL and title of the web pages.
4. Record the relevant details in the observation table.

STEP 4: Run Chrome CacheView

1. The tool displays the files stored in the Chrome cache.
2. Observe the URL, content type, file size and last accessed time.
3. Record the cache information for analysis.

STEP 5: Analyze Browser Artifacts

1. Examine the history and cache entries.
2. Compare the visit times and accessed times.
3. Identify the available browser artifacts.
4. Document the findings for forensic investigation.

STEP 6: Record the Observation

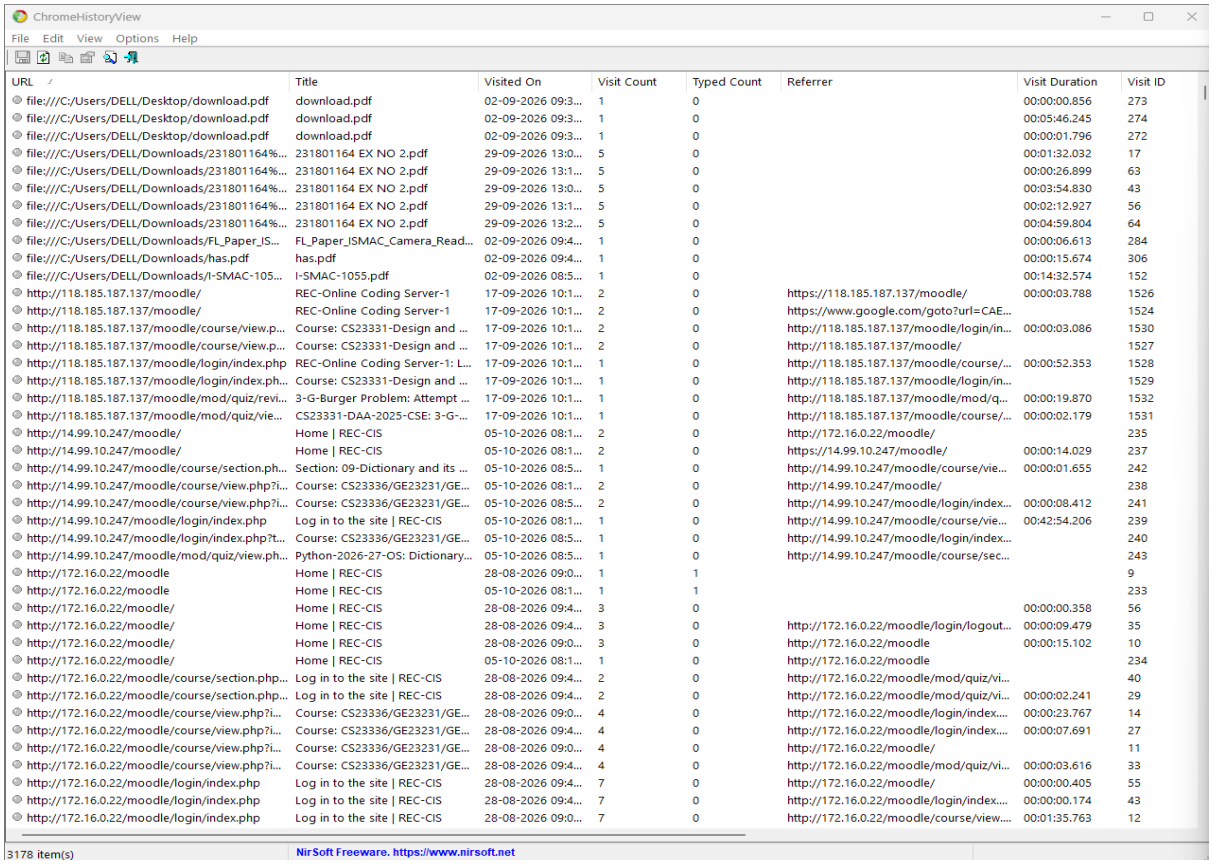
OBSERVATION:

Browser History

S.No.	URL	Title	Visit Date/Time
1	https://www.google.com	Google	15-09-2026, 4:10 PM
2	https://www.youtube.com	YouTube	15-09-2026, 4:30 PM

Browser Cache

S.No.	Details	Observation
1	URL	https://google.com
2	Content Type	Text/HTML
3	File Size	12 KB
4	Last Accessed Time	15-09-2026, 9:10 AM
5	Server Name	www.google.com




```
==================================================
URL                : file:///C:/Users/DELL/Desktop/download.pdf
Title              : download.pdf
Visited On         : 02-09-2026 09:34:47
Visit Count        : 1
Typed Count        : 0
Referrer           :
Visit Duration     : 00:00:00.856
Visit ID           : 273
Profile            : Default
URL Length         : 42
Transition Type    : Reload
Transition Qualifiers: Chain Start,Chain End
History File       : C:\Users\DELL\AppData\Local\Google\Chrome\User Data\Default\History
==================================================

==================================================
URL                : file:///C:/Users/DELL/Desktop/download.pdf
Title              : download.pdf
Visited On         : 02-09-2026 09:34:48
Visit Count        : 1
Typed Count        : 0
Referrer           :
Visit Duration     : 00:05:46.245
Visit ID           : 274
Profile            : Default
URL Length         : 42
Transition Type    : Reload
Transition Qualifiers: Chain Start,Chain End
History File       : C:\Users\DELL\AppData\Local\Google\Chrome\User Data\Default\History
==================================================
```

RESULT:

Thus, browser artifacts such as history and cache were extracted and analyzed using forensic tools successfully.

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DATE: 15.9.26 ROLL No.:241901109

EXPERIMENT 7

COMPARISON OF TWO FILES FOR FORENSIC INVESTIGATION USING COMPARE IT SOFTWARE

AIM:

To compare two files using Compare It software and identify differences through side-by-side comparison and highlighted changes for forensic investigation.

PROCEDURE:

STEP 1: Prepare Files

1. Create two files using Notepad: Report1.txt and Report2.txt.
2. Create similar content with some differences and save both files.

STEP 2: Open Compare It

1. Install and launch Compare It software.
2. Select the Compare It tool to open the file selection window.

STEP 3: Select Files

1. Select Report1.txt and click Open.
2. Select Report2.txt and click Open.
3. Both files will be displayed side by side.

STEP 4: Compare Files

1. Compare the contents of both files.
2. Observe the highlighted differences, including added, deleted and modified text.
3. Record the identified changes.

STEP 5: Generate Report

1. Open the print or report option.
2. Generate a coloured difference report.
3. Save or print the report for forensic documentation.

OBSERVATION:

S.No.	File Name	Observation	Result
1	Report1.txt	Original file loaded successfully	File opened
2	Report2.txt	Modified file loaded successfully	File opened
3	Report1.txt & Report2.txt	Both files displayed side by side	Comparison completed
4	Line 4	Status changed from Pending to Completed	Difference detected
5	Line 5	File size changed from 1024 KB to 2048 KB	Difference detected
6	Comparison View	Changed text highlighted with colours	Changes identified
7	Print Report	Difference report generated	Report obtained

```
FORENSIC INVESTIGATION REPORT

Case ID: 102
Case Type: Cyber Evidence Analysis
Investigator: Karthika
Location: Avadi
Date: 09-10-2026
Time: 03:45 PM
Evidence ID: EVD-002
Evidence Type: Document File
Device: Laptop Computer
Operating System: Windows 10
Evidence Status: Under Investigation

DESCRIPTION OF EVIDENCE

The evidence file was collected from the suspect system.
The file was preserved for detailed forensic examination.
The contents were analyzed using investigation software.
Several changes were observed during the examination.

INVESTIGATION DETAILS

The investigator examined the document contents.
The file contained information related to the incident.
The original document was compared with a modified copy.
Multiple differences were identified during the comparison.

FINDINGS

File size: 5 KB
File format: DOC
File owner: karthika
File location: Documents
Security status: Suspicious

CONCLUSION

The evidence was successfully analyzed.
The comparison revealed multiple differences between the two files.
The findings were documented for further investigation.
```

```
FORENSIC INVESTIGATION REPORT

Case ID: 101
Case Type: Digital Evidence Analysis
Investigator: Keerthika k
Location: Chennai
Date: 08-10-2026
Time: 03:35 PM
Evidence ID: EVD-001
Evidence Type: Text File
Device: Desktop Computer
Operating System: Windows 11
Evidence Status: Active

DESCRIPTION OF EVIDENCE

The evidence file was collected from the investigation system.
The file was preserved for forensic examination.
The contents were analyzed using forensic tools.
No modifications were made during the initial examination.

INVESTIGATION DETAILS

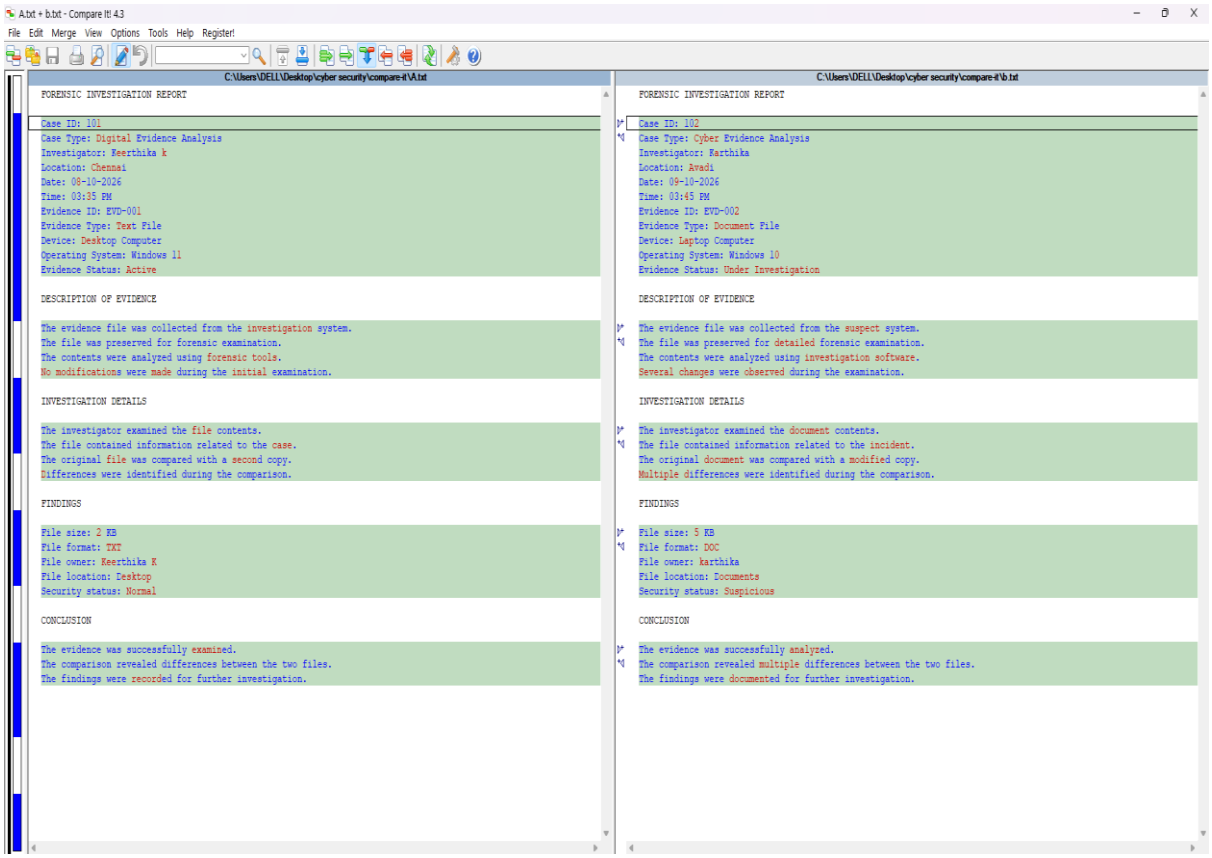
The investigator examined the file contents.
The file contained information related to the case.
The original file was compared with a second copy.
Differences were identified during the comparison.

FINDINGS

File size: 2 KB
File format: TXT
File owner: Keerthika K
File location: Desktop
Security status: Normal

CONCLUSION

The evidence was successfully examined.
The comparison revealed differences between the two files.
The findings were recorded for further investigation.
```



RESULT:

Thus, the two files were successfully compared using Compare It software, and the differences were identified through highlighted changes and a comparison report.

NAME: SIVARANJANII. M

DATE: 22.9.26 ROLL No.:241901109

EXPERIMENT 8

COLLECTING EMAIL EVIDENCE IN VICTIM PC

AIM:

To collect email evidence from a victim PC by capturing RAM and extracting email IDs using DumpIt and Bulk Extractor Viewer.

PROCEDURE:

STEP 1: Capture RAM

1. Run the DumpIt.exe tool on the authorized investigation computer.
2. The tool generates a physical memory dump of the Windows system.
3. Enter Y when prompted to start the process.
4. Save the generated RAW memory dump file in the specified directory.

STEP 2: Open Bulk Extractor Viewer

1. Download and install Bulk Extractor Viewer.
2. Launch the software.
3. Select the option to generate a report.

STEP 3: Select Memory Dump

1. Browse and select the RAW memory dump file created using DumpIt.
2. Choose an output folder to save the report.
3. Start the Bulk Extractor analysis process.

STEP 4: Analyze Email Evidence

1. Wait for the report generation to complete.
2. Open the generated report in Bulk Extractor Viewer.
3. Select email.txt to view the extracted email addresses.
4. Open email_histogram.txt to examine the email address frequency information.
5. Record the relevant observations in the lab record.

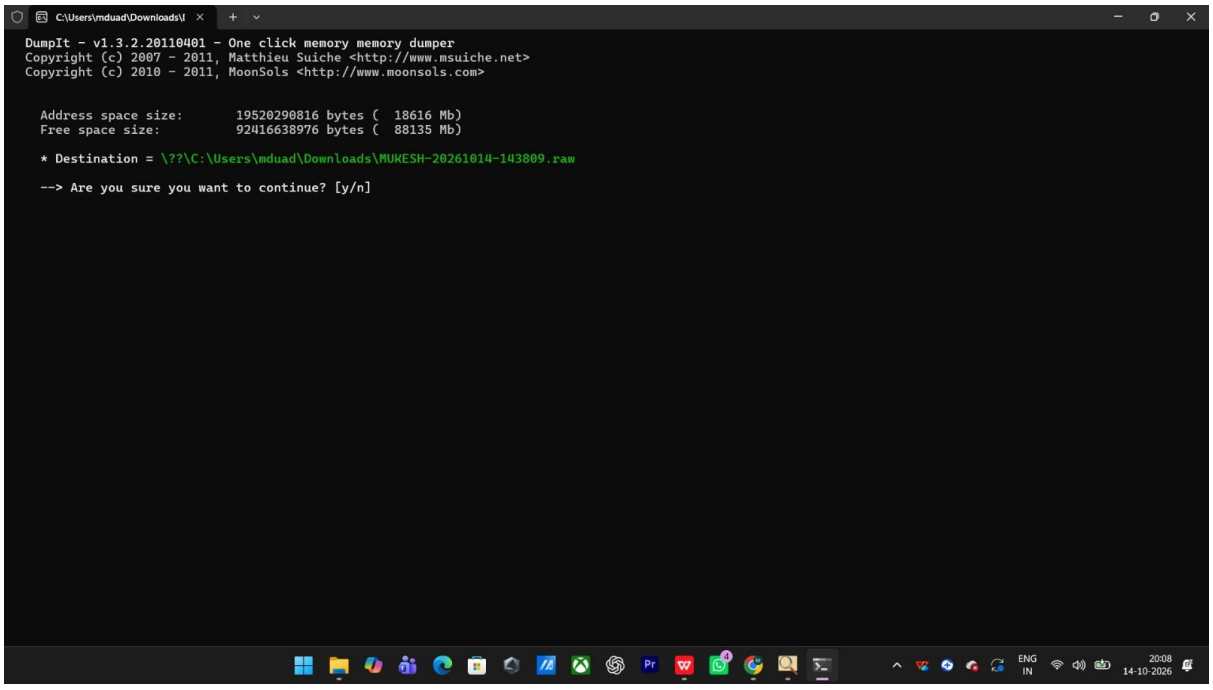
OBSERVATION:

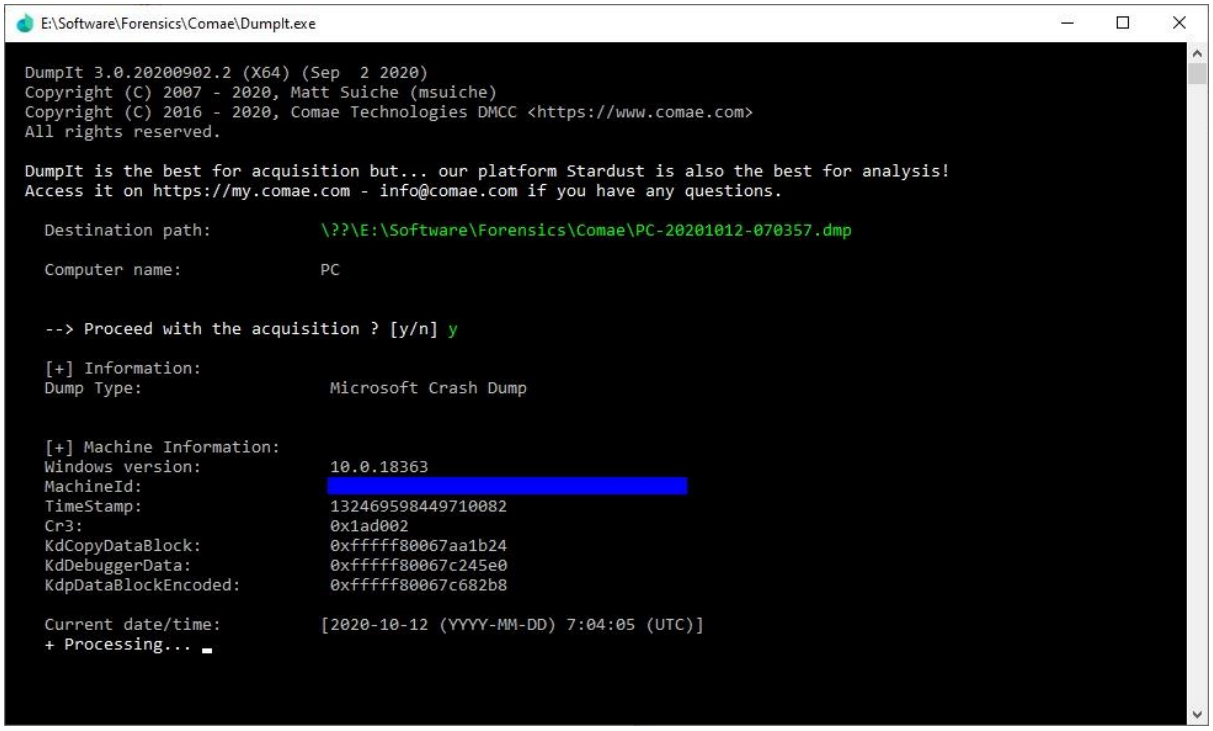
Email Evidence Table

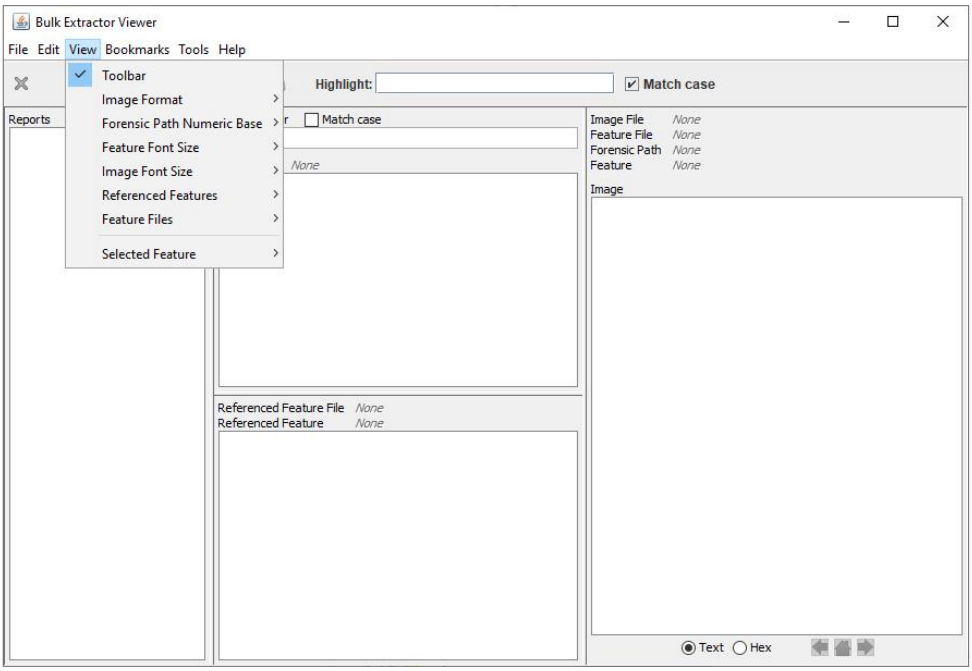
S.No.	Extracted Email ID	Source File	Observation
1	sivaranjanii@gmail.com	email.txt	Email address identified
2	admin@rec.org	email.txt	Email address identified
3	support@rec.net	email.txt	Email address identified

Email Histogram Table

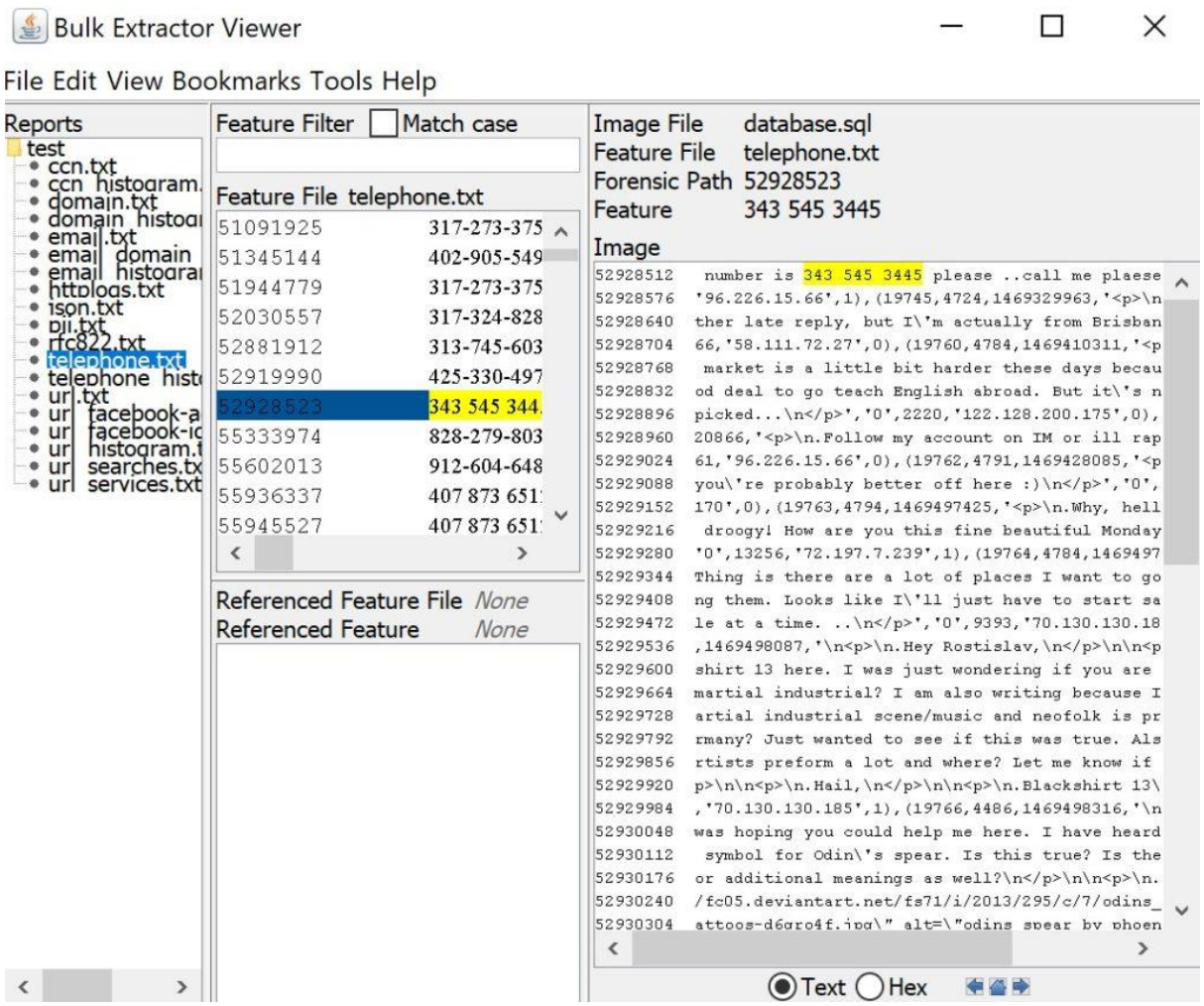
S.No.	Email ID	Frequency
1	sivaranjanii@gmail.com	5
2	admin@rec.org	3
3	support@rec.net	2







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RESULT:

Thus, the RAM dump was collected using DumpIt, and email evidence was extracted and analyzed using Bulk Extractor Viewer.

NAME: SIVARANJANII. M

DATE: 29.9.26 ROLL No.:241901109

EXPERIMENT-9

STUDY THE STEPS FOR HIDING AND EXTRACT ANY TEXT FILE BEHIND AN IMAGE FILE/ AUDIO FILE USING COMMAND PROMPT

AIM:

To hide a text file behind an image or audio file and extract it using Command Prompt.

PROCEDURE:

STEP 1: Prepare Files

1. Create a text file named secret.txt.
2. Select an image file named image.jpg or an audio file named audio.mp3.
3. Place both files in the same folder.

STEP 2: Open Command Prompt

1. Open the folder containing the files.
2. Click the address bar, type cmd, and press Enter.

STEP 3: Hide the Text File Behind an Image

1. Enter the following command:

```
copy /b image.jpg + secret.txt output.jpg
```

2. The combined file output.jpg will be created.

STEP 4: Extract the Text File

1. Open Command Prompt in the same folder.
2. Enter the following command:

```
findstr /V /R "^$" secret.txt
```

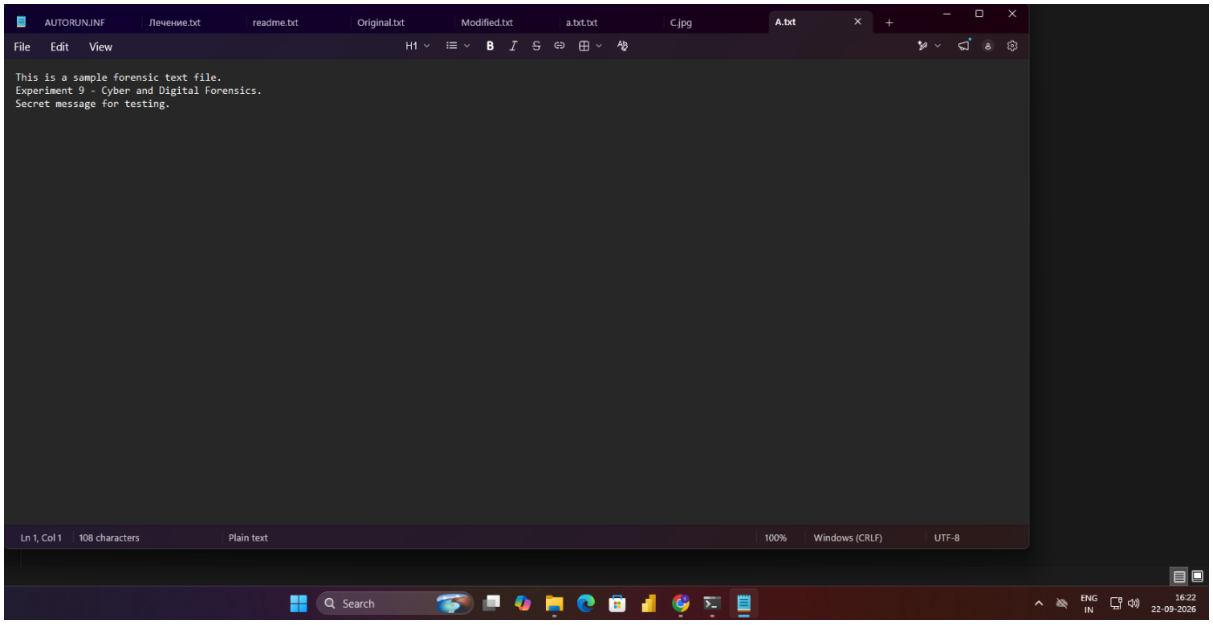
This command displays the text file's contents, but it does not extract the hidden text from output.jpg. For a simple demonstration, use the original secret.txt file to verify its contents.

STEP 5: Verify the Output

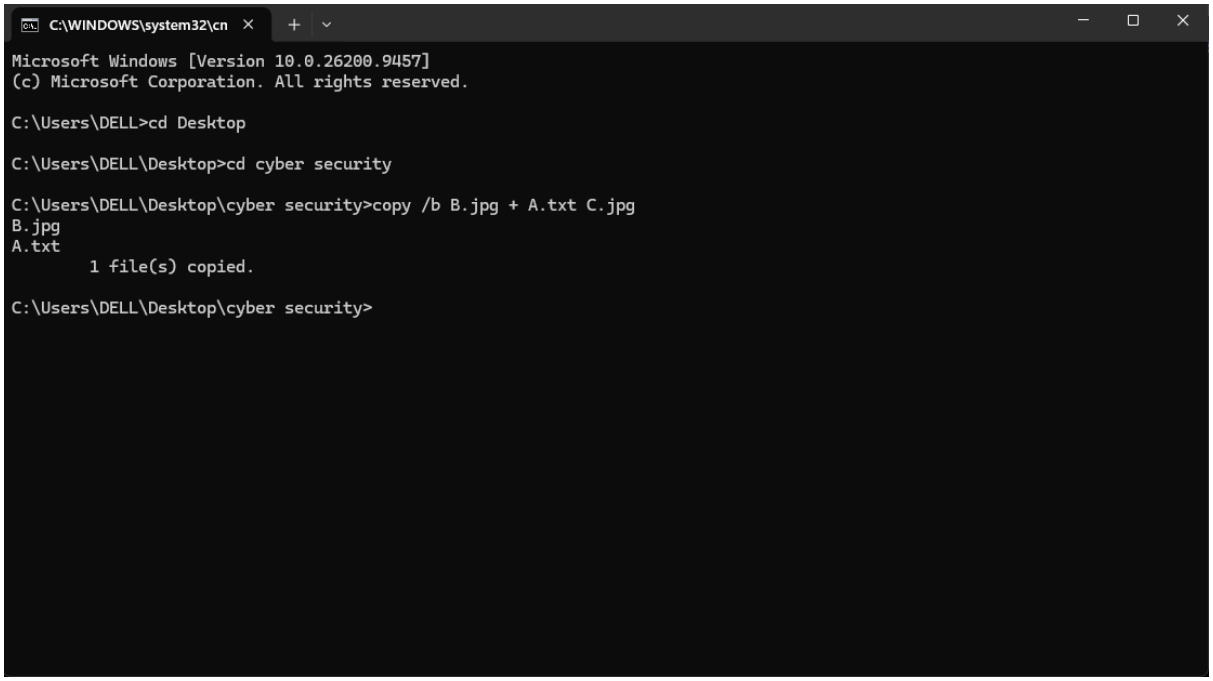
1. Open output.jpg to check whether the image is still viewable.
2. Check that the combined file has been created successfully.

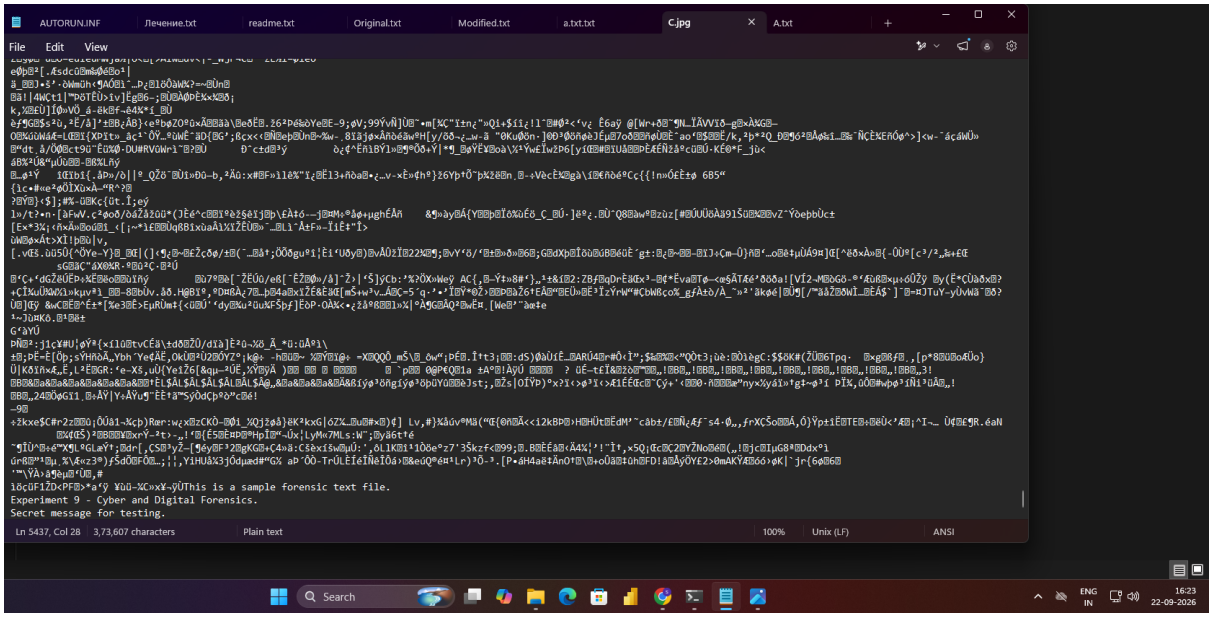
OBSERVATION:

S.No.	Details	Observation
1	Original Image File	image.jpg
2	Text File	secret.txt
3	Command Used	copy /b image.jpg + secret.txt output.jpg
4	Output File	output.jpg
5	Result	Image and text data combined successfully









RESULT:

Thus, a text file was combined with an image file using Command Prompt, demonstrating a basic file-concealment technique.

NAME: SIVARANJANII. M

DATE: 6.10.26 ROLL No.:241901109

EXPERIMENT-10

EXTRACT EXCHANGEABLE IMAGE FILE FORMAT (EXIF) DATA FROM IMAGE FILES USING EXIFREADER SOFTWARE

AIM:

To extract and analyze EXIF data from image files using ExifReader software.

PROCEDURE:

STEP 1: Open ExifReader

1. Download and open ExifReader software.
2. Launch the software on the computer.

STEP 2: Select an Image File

1. Choose an image file from the computer.
2. Open the selected image in ExifReader.

STEP 3: Extract EXIF Data

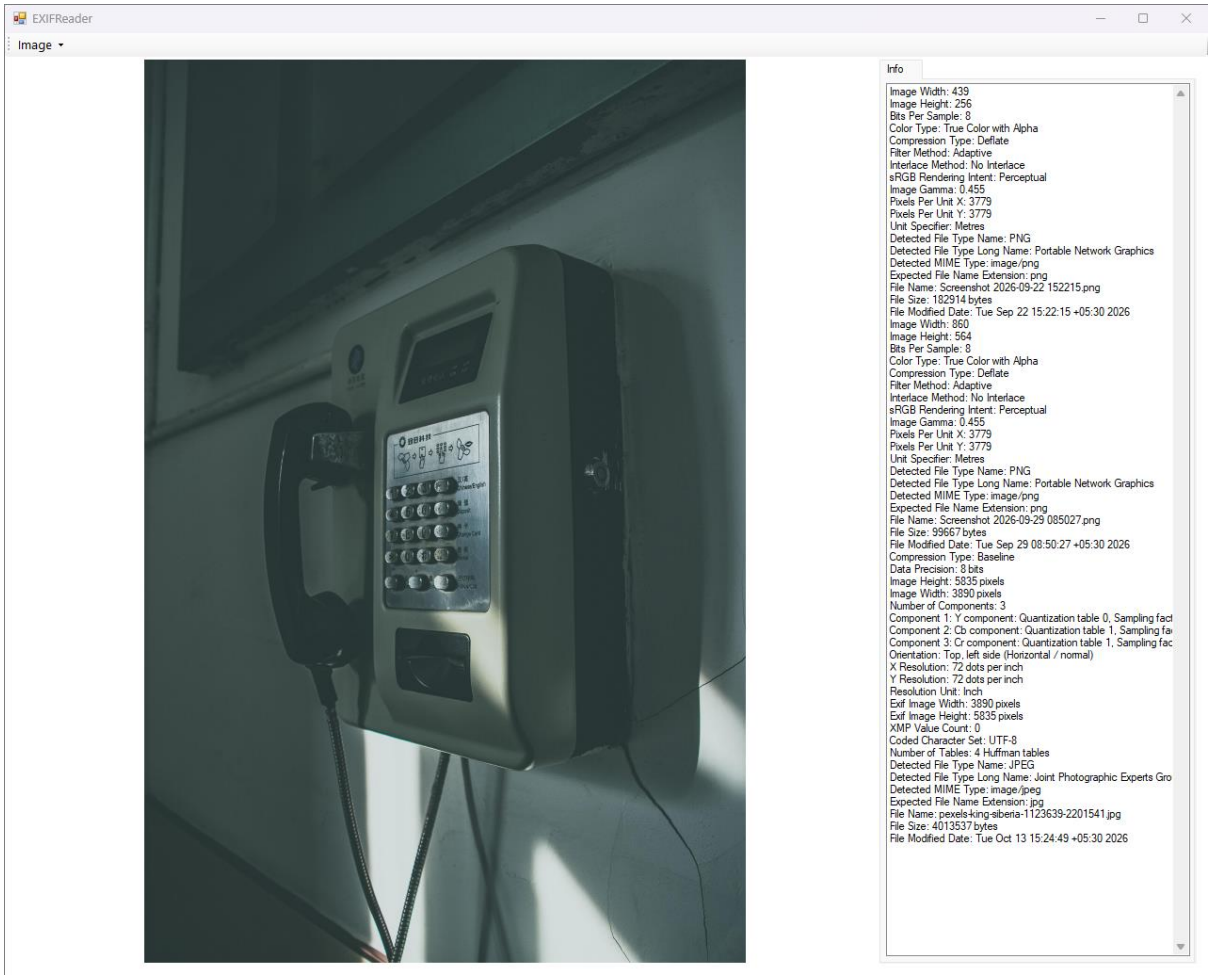
1. Allow the software to read the image metadata.
2. Observe the EXIF details displayed by the software.
3. Check the camera model, date and time, image dimensions, and other available details.

STEP 4: Analyze EXIF Data

1. Examine the extracted metadata.
2. Identify the available image information.
3. Record the relevant details in the observation table.

OBSERVATION:

S.No.	Details	Observation
1	File Name	Sample.jpg
2	File Type	JPEG Image
3	Image Dimensions	1920 × 1080 pixels
4	Camera Model	Available in metadata
5	Date and Time	Available in metadata
6	EXIF Data	Extracted successfully



RESULT:

Thus, EXIF data was extracted and analyzed from an image file using ExifReader software successfully.